

Day 19 - Lecture 19.6: Stack Plots (Area Charts)

1. Overview & Purpose

A **Stack Plot** (also called a Stacked Area Chart) displays multiple data series stacked on top of each other along a continuous or sequential x-axis (such as time, days, or months).

Why use a Stack Plot?

- **Part-to-Whole over Time:** While a **Pie Chart** shows part-to-whole for a single static point in time, a **Stack Plot** shows how both the individual parts and the cumulative total evolve over time.
 - **Contribution Analysis:** Easily observe which sub-category is growing or shrinking in proportion to the total.
 - **Total Magnitude:** The topmost line shows the cumulative total across all series.
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2. Syntax & Parameters

```
plt.stackplot(  
    x,                                # Sequential x-axis (e.g., days, years)  
    y1, y2, y3, ...,                 # Multiple 1D arrays of equal length representing lay  
    labels=['L1', 'L2'],              # Labels for each category  
    colors=['c1', 'c2'],              # Fill colors for each layer  
    alpha=0.8,                        # Transparency  
    baseline='zero'                   # 'zero' (standard), 'sym' (streamgraph), or 'wiggle'  
)
```

3. Real-World Case Study: Website Traffic Channels

Tracking visitor traffic across days of the week:

- **Direct Traffic:** Users typing the URL directly.
- **Organic Traffic:** Users arriving via search engines (Google, Bing).
- **Social Traffic:** Users arriving from LinkedIn, Twitter, YouTube.

```
import matplotlib.pyplot as plt

days = ["Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun"]
direct = [50, 60, 70, 80, 90, 100, 110]
organic = [30, 40, 50, 55, 60, 70, 80]
social = [20, 25, 30, 35, 40, 50, 60]

plt.figure(figsize=(9, 6))
plt.stackplot(days, direct, organic, social, labels=['Direct Traffic', 'Organic Search', 'Social Media'],
              colors=['#4C72B0', '#55A868', '#C44E52'], alpha=0.85)

plt.title("Website Traffic Channels Over a Week", fontsize=14, fontweight="bold")
plt.xlabel("Day of the Week", fontsize=12)
plt.ylabel("Number of Visitors", fontsize=12)
plt.legend(loc='upper left', fontsize=11)
plt.grid(True, linestyle=":", alpha=0.5)
plt.show()
```

4. Key Takeaways & Limitations

- **Readability Caution:** Because layers are stacked on top of each other, only the bottom-most layer has a flat baseline. Higher layers have curved baselines, making it harder to judge exact numerical values of upper layers.
- Limit the number of categories to 3–5. Beyond that, the chart becomes cluttered and hard to interpret.